

Trustworthy fall detection: A responsible AI approach for real-time video monitoring of the elderly

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ABSTRACT

Falls among older adults represent a serious public health problem, particularly for individuals living alone or in assisted-care environments, where delayed intervention can lead to severe injuries or death. Although numerous fall-detection systems have been proposed, many existing approaches suffer from limitations, including high computational complexity, limited interpretability, reliance on centralized video storage, and reduced generalization across diverse environments. These challenges motivate the development of lightweight, explainable, and trustworthy fall-detection framework for healthcare applications. To address these limitations, this paper proposes a vision-based fall detection framework that integrates MobileNetV2 for efficient spatial feature extraction and a Gated Recurrent Unit (GRU) for temporal motion modeling. In addition, a federated learning strategy is employed to support collaborative training across distributed data sources without requiring centralized collection of raw video recordings, thereby reducing reliance on centralized storage while enabling learning from heterogeneous environments. During deployment, a sliding-window inference strategy is adopted to achieve reliable and low-latency fall detection. Model transparency is further supported through Grad-CAM++ visual explanations, while Fairlearn is used to assess performance fairness across different fall types and user groups. Experimental results on the combined CAUCAFall and Le2i datasets demonstrate that the proposed framework achieves an accuracy of 97.14%, a precision of 99.26%, a recall of 96.75%, and an F1-score of 97.99%. Although the achieved accuracy is 2.46 percentage points lower than the highest reported accuracy (99.60%), the competing method employs a substantially more complex multi-stage architecture. In contrast, the proposed framework adopts a lightweight MobileNetV2-GRU architecture while integrating explainability through Grad-CAM++, fairness evaluation using Fairlearn, and decentralized collaborative learning. Furthermore, the proposed framework achieves an inference throughput of approximately 93 FPS on the experimental platform used in this study, demonstrating high computational efficiency.

1. Introduction

Falls among older adults represent a serious public health concern, especially for elderly individuals who live alone or reside in assisted-care and nursing facilities. Many older adults prefer to remain independent by living in their own homes, while others rely on institutional care due to age-related health conditions. In both situations, regular monitoring is crucial, as older adults are more susceptible to falls and related injuries. However, providing constant manual supervision for every individual is often impractical, costly, and time-consuming, particularly in long-term care settings. When assistance is delayed after a fall, the risk of severe injury, long-term disability, or death increases significantly. This issue is further compounded by the high

frequency of falls among older adults worldwide, making it not only an individual safety concern but also a major public health problem (Giovannini et al., 2022). Falls are common and recurring events in this age group and are recognized as the sixth leading cause of morbidity and mortality among older adults (Chen et al., 2025; El Sayed et al., 2023). According to the World Health Organization (WHO), more than 500,000 fatal falls occur globally each year, most of which involve individuals aged 65 years and older (World Health Organization, 2021). Additionally, nearly one-third of older adults experience at least one fall each year, and about 10% of these incidents result in serious injuries, such as fractures or intracranial trauma (Kakara et al., 2025). These statistics underscore the significant health burden posed by falls among the elderly population (Kalache et al., 2007; Yang et al., 2025).

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Given the serious impact of falls on the health and safety of older adults, rapid detection and assistance are essential to reduce the risk of severe injuries. Automated monitoring systems provide a practical solution by continuously observing individuals, and promptly sending alerts to family members or nursing staff in case of detecting fall incidents. This rapid response can significantly improve outcomes by ensuring that help is provided as quickly as possible. To achieve this goal, researchers have developed various fall-detection approaches. Some methods rely on wearable sensors (Yao et al., 2026), while others use vision-based techniques to monitor individuals and enhance their safety without requiring constant human supervision (Apicella & Snidaro, 2021; Batumalai et al., 2025; Liu et al., 2025). Wearable sensor methods, particularly those employing inertial measurement units (IMUs), have attracted considerable research interest due to their compact size, low cost, and high detection accuracy (Krupitzer et al., 2019). Despite these advantages, their adoption among older adults remains limited, as such devices often cause discomfort and require frequent recharging of batteries, which can negatively affect user compliance. Moreover, wearable sensors are susceptible to various sources of noise, including improper device placement and dynamic interactions between the body and the sensor, which can lead to increased false alarm rates. In addition, these systems are often associated with higher installation and maintenance costs and are sensitive to environmental conditions and interference (Fei et al., 2023; Wickramasinghe et al., 2017).

By contrast, computer vision-based fall-detection methods overcome many of the limitations of wearable systems by enabling non-intrusive monitoring without requiring older adults to wear specialized devices. By analyzing video data, these approaches can capture rich contextual information about body posture and movement, which supports more reliable fall detection in real-world settings (Yu et al., 2025). For this reason, video-based non-contact fall detection has attracted increasing attention as an effective alternative, offering continuous real-time monitoring without active user involvement (Chen et al., 2024; Huang et al., 2024; Liu et al., 2025; Newaz & Hanada, 2024; Newaz et al., 2023; Yang et al., 2024). This is particularly important for older adults with cognitive impairments, such as dementia, for whom wearing and maintaining sensor-based devices may be difficult. In addition, the widespread availability of cameras allows vision-based systems to take advantage of existing surveillance infrastructure, enabling scalable deployment not only in smart homes but also in hospitals, nursing facilities, and other public or institutional environments. At the same time, recent advances in deep learning have significantly improved the accuracy of vision-based models by enhancing their ability to learn spatiotemporal patterns from video data. As a result, the performance gap between camera-based and sensor-based fall-detection systems have been substantially reduced (Swarubini et al., 2025).

Building on these advances, many vision-based fall-detection methods have been proposed. These methods use different learning strategies to analyze video data and detect fall events. Existing approaches can be broadly divided into several categories. Some methods use end-to-end spatio-temporal models, such as 3D convolutional networks or transformer-based architectures, to directly learn motion patterns from video sequences. Other approaches rely on object detection frameworks, often based on YOLO models, to localize the human body, after which postural changes are inferred from bounding-box geometry, motion patterns, or additional pose estimation modules to indicate a fall. Another group of methods focuses on pose or skeleton-based analysis, where body joint trajectories are modeled using recurrent networks, graph convolutional networks, or attention mechanisms.

Despite promising results on benchmark datasets, many of these methods still present important limitations. Several approaches rely on complex multi-stage pipelines or computationally intensive architectures, resulting in increased computational cost and reduced computational efficiency. Other methods depend on centralized training with sensitive video data, increasing reliance on centralized storage and

motivating the need for decentralized learning frameworks. In addition, many existing models lack interpretability and fairness analysis, making their decisions difficult to understand and potentially biased in healthcare applications. These limitations motivate the development of lightweight, explainable, fair, and decentralized fall-detection frameworks that support collaborative training while reducing the need for centralized storage of raw video data.

Compared with existing fall-detection approaches, this paper proposes an integrated algorithm for reliable and trustworthy elderly fall detection that directly addresses several limitations of prior work. Rather than relying on multi-stage and computationally demanding pipelines involving separate human detection, pose estimation, or complex transformer-based models, the proposed framework adopts a lightweight end-to-end design based on MobileNetV2 (Sandler et al., 2018) and a GRU network. This design enables accurate and computationally efficient fall detection while maintaining a lightweight architecture. Unlike approaches that rely on fixed-length temporal segmentation or short-term motion cues, the proposed GRU-based temporal modeling captures motion dynamics across consecutive frames, improving robustness to slow and gradual falls. Furthermore, while many existing methods depend on centralized training and access to raw video data, the proposed framework incorporates federated learning to support collaborative model training across distributed data sources through model aggregation rather than raw data exchange. This decentralized training strategy reduces reliance on centralized storage and supports deployment across diverse healthcare and assisted-living environments. To further promote trust and transparency, the proposed method incorporates Grad-CAM++ (Chattopadhyay et al., 2018) to provide interpretable visual explanations of model decisions, allowing caregivers and practitioners to understand which discriminative image areas influence fall detection outcomes.

Moreover, a fairness evaluation is conducted to assess performance consistency across different fall scenarios and user groups, addressing ethical considerations that are often overlooked in existing fall-detection systems. Overall, this work presents a lightweight, explainable, and trustworthy fall-detection framework for elderly monitoring. By combining efficient spatial and temporal modeling with decentralized training, explainability, and fairness assessment, the proposed framework addresses key challenges related to computational complexity, centralized data management, interpretability, and equitable performance. Consequently, it provides a promising approach for healthcare and assisted-living applications.

Accordingly, the main contributions of this paper are summarized as follows:

1. A lightweight fall detection framework is proposed that combines MobileNetV2 for efficient spatial feature extraction with a GRU network for capturing temporal motion dynamics, enabling accurate and computationally efficient fall recognition.
2. A decentralized federated learning strategy is introduced to enable collaborative model training across multiple data sources without transferring raw video data, reducing storage burden and supporting model generalization across diverse environments.
3. An explainability component using Grad-CAM++ is incorporated, which provides visual insights into the spatial regions influencing the model's decisions, improving interpretability, trustworthiness, and transparency.
4. A fairness evaluation using Fairlearn is conducted to assess potential performance disparities across different subgroups, supporting responsible, unbiased, and equitable AI deployment.
5. A practical sliding-window inference mechanism is developed to improve computational efficiency while preserving high detection accuracy.
6. Comprehensive experiments are evaluated on multi-view datasets to validate the effectiveness, robustness, and ethical alignment of the proposed fall detection framework.

2. Related work

Existing fall-detection research can be broadly categorized based on the sensing modality and modeling strategy used. Early studies explored sensor-based approaches, while more recent work has focused on vision-based methods, driven by advances in deep learning and the widespread availability of cameras.

2.1. Sensor-based fall detection

Some studies relied on non-visual sensors to detect falls. For example, [Patel et al. \(2026\)](#) proposed a human fall detection system that leveraged ultrasonic sensors and a hybrid deep learning model. The authors integrated the Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BLSTM) to classify time-series sensor data. This system was embedded within a Space-Air-Ground Integrated Network (SAGIN) framework, enabling real-time alerts. Similarly, [Khan et al. \(2025\)](#) proposed an explainable machine-learning-based fall detection and alert system that integrated smartphone sensor data with a cloud-based framework. This approach used inertial data captured from smartphone accelerometers and gyroscopes processed through a Gradient Boosted Decision Tree (GBDT) model to classify falls. Also, the method provided explainability via SHapley Additive exPlanations (SHAP) to clarify how individual features influence predictions. While such sensor-based approaches could achieve good accuracy under controlled conditions, their real-world deployment remains challenging due to device availability, user compliance, battery requirements, and sensitivity to noise, limitations that are particularly pronounced among older adults.

2.2. Vision-based methods

To overcome the limitations of wearable sensors, more recent studies have focused on vision-based fall detection, enabled by advances in deep learning and the widespread availability of cameras. These methods analyze video data and do not require users to wear any devices.

2.2.1. Appearance features and spatio-temporal modeling methods

Many vision-based approaches rely on appearance features and motion patterns extracted from RGB video. Several methods used object detection frameworks, particularly YOLO-based models, to detect and track individuals and infer falls from changes in body orientation and bounding-box motion. [Zheng et al. \(2024\)](#) proposed an improved YOLOv8 framework in which the C2f module in the backbone network is replaced with the FasterNet module, while a deformable convolution module was introduced in the neck section, and a triplet attention mechanism was incorporated during multi-scale feature fusion. On the other hand, [Khekan et al. \(2025\)](#) proposed a human fall detection framework using an improved YOLOv8 model. The model was trained and evaluated on the CAUCAFall dataset, which comprised video sequences representing various fall types and daily activities in unconstrained indoor environments. Furthermore, [Chen et al. \(2022\)](#) proposed an elderly fall-detection model based on YOLOv5s. The method replaced standard 3×3 convolutions with Asymmetric Convolution Blocks (ACB) to better extract horizontal, vertical, and rotational features. It integrated a spatial attention module (SAM) to focus on important positional cues, and removed the small-scale 76×76 detection layer to prioritize richer semantic features. In addition, [Zaghden et al. \(2025\)](#) introduced a deep learning-based fall detection framework that enhances YOLOv8 with attention mechanisms to improve feature representation and detection accuracy. Furthermore, [Sanjalawe et al. \(2025\)](#) proposed a video-based system combining YOLOv8 for spatial human detection and Time-Space Transformers (TST) for temporal motion analysis. Using the CUCAFall and DiverseFALL10500 datasets, the model detects falls by integrating YOLOv8's real-time object detection with TST's ability to capture motion across frames. Although these methods are effective,

they are often sensitive to occlusions, camera viewpoint changes, and complex backgrounds.

Other studies directly learn spatio-temporal patterns from video sequences. In [Núñez-Marcos and Arganda-Carreras \(2024\)](#), Marcos et al. proposed a vision-based fall detection framework that leveraged a transformer, specifically a Uniformer backbone combining self-attention with efficient computation to directly learn fall patterns from raw RGB video frames. The model processed short video clips in a sliding-window to classify whether a fall had occurred. The authors used two publicly available benchmark datasets; UP-Fall dataset and the UR Fall dataset. Similarly, [Alam et al. \(2025\)](#) proposed a vision-based human fall detection system using transfer learning with a 3D CNN to capture spatio-temporal information from video sequences. A pretrained 3D CNN model (trained on Sports-1M) is used as a fixed feature extractor, where input videos are segmented into 16-frame clips through the network to extract features; these features are then classified using a support vector machine (SVM). In a related study, [Zou et al. \(2021\)](#) introduced a spatio-temporal fall-detection approach that analyzed spatial localization and temporal motion patterns. A 3D CNN is integrated with an SSD-based object detection framework to generate constrained movement tubes, which track a person through a sequence of refined bounding boxes and help mitigate background interference. The 3D CNN encodes both appearance and motion cues, while additional layers perform tube anchor generation, classification, and regression. In [Fu \(2025\)](#), Fangping Fu proposed a vision-based fall detection system that exploited 3D sparse convolutional neural networks (3D-SCNN). A dynamic key-frame selection preprocessing stage is incorporated using jitter buffers to adaptively select salient frames. To preserve feature continuity before fed to the sparse 3D-CNN model, overlapping spatio-temporal cubes are constructed. An alternative approach presented by [Gaya-Morey et al. \(2023\)](#) proposed an explainable deep learning framework for human activity recognition tailored to elderly people. This approach leveraged backbone models such as TimeSformer, TANet, and TPN for video-based HAR and integrated explainable AI (XAI) techniques to generate space and time-aligned explanations that make decisions more understandable to caregivers and end users.

Despite their strong performance on benchmark datasets, these methods often require high computational resources. Their accuracy can also decrease in real-world conditions due to occlusions, lighting changes, cluttered backgrounds, and camera viewpoint variations. These factors make real-time deployment on low-power devices challenging.

2.2.2. Pose and skeleton-based approaches

Another major research direction focuses on human pose or skeleton analysis, where fall detection is performed by modeling body joint trajectories. [Romaissa et al. \(2020\)](#) addressed elderly fall detection by leveraging human body geometry extracted from video sequences, where angular and distance features are calculated between the head centroid and the body's hip center. These features are used to train both a two-class SVM and an LSTM network for distinguishing falls from normal activities. Related work by [Bach et al. \(2026\)](#) proposed a vision-based fall detection method based on YOLOv8n-pose for human pose estimation, it incorporated Dynamic Cross-Stage Partial Fusion (DyC2F) blocks to enhance feature representation. The temporal dynamics of human motion are captured using an LSTM module that processes keypoint trajectories. Whereas, [Kong et al. \(2025\)](#) presented a vision-based fall detection system that combined deep learning-based pose estimation with geometric feature analysis. The method applied the YOLOv11 object- and pose-detection model to extract 17 key body landmarks from video frames, it computed nine geometric pose angle features (e.g., center of mass and joint angles), and trained an SVM classifier to detect transitions from upright to lying posture. On the other hand, [Tran et al. \(2025\)](#) proposed a fall-detection system that combined MediaPipe key-point extraction and either an LSTM network or a K-Nearest Neighbor (KNN) classifier. The method identifies human falls

in real time by analyzing skeletal motion patterns derived from key-point sequences.

Tirziu et al. (2025) proposed a vision-based fall detection system that integrated the YOLOv11 object detection and pose estimation model with an AI-based voice interaction module. The system was analyzed, including video input from a dual-camera setup, skeletal keypoints for posture evaluation, and falls based on dynamic changes in body pose. It engaged a voice assistant to verify the user's condition and reduce false alarms. An alternative framework is presented by McCall et al. (2024) for identifying and forecasting fall risks utilizing a transformer model trained on 2D poses extracted through an off-the-shelf pose extractor, incorporating transfer learning techniques. Initially, the transformer is trained on a large dataset containing 2D poses of general actions. Subsequently, the method froze the majority of its layers and fine-tuned only the last few layers using relatively smaller datasets for fall detection and prediction tasks. Gao et al. (2023) proposed a fall-detection approach that integrated human pose estimation with a compact convolutional architecture. The method first extracted skeletal keypoints using OpenPose and labeled them in the input images. These keypoints are then combined with the original visual information and fed into a modified MobileNetV2 network to classify fall and non-fall events. However, Wang and Lin (2024) presented a camera-based monitoring system that protects privacy by replacing elderly individuals' images with 2D avatars while enabling accurate fall detection. Using YOLOv8 for real-time person detection and pose estimation, combined with a residual causal CNN for motion analysis.

In addition, several approaches perform fall detection using only joint or skeleton representations, without processing full human images, thereby enhancing privacy by minimizing the exposure of identifiable visual details. Abdo et al. (2023) proposed a three-stage deep learning framework for vision-based fall detection. This method combined YOLO for human detection, OpenPose for skeleton extraction, and a Spatial-Temporal Graph Convolutional Network (ST-GCN) for classifying fall vs. non-fall events. Whereas, Castro et al. (2023) proposed two vision-based fall-detection systems that use 2D and 3D human pose estimation combined with an LSTM network enhanced by an attention mechanism. Using OpenPose (2D) and FrankMocap (3D), the authors extract body-joint coordinates from video frames. While pose-based methods offer interpretability and structured motion representation, their performance heavily depends on accurate key-point detection and degrades under occlusions, poor lighting, or partial body visibility.

Pose-based methods offer better privacy and clearer interpretation of body motion, as they do not rely on full human images. However, their performance strongly depends on accurate pose estimation. Occlusions, poor lighting, and partial body visibility can lead to missing or incorrect joint detection, which reduces detection accuracy and robustness.

3. Proposed method

In this work, we introduce a lightweight fall detection framework that combines MobileNetV2 for spatial feature extraction, a GRU network for temporal motion modeling, and Grad-CAM++ for model explainability. The framework further incorporates a federated learning strategy to enable decentralized training across distributed data sources without sharing raw video data. MobileNetV2 generates compact spatial representations with high computational efficiency, while the GRU captures motion dynamics across consecutive frames to accurately distinguish falls from daily activities. The federated learning framework supports collaborative model training across multiple camera sources while reducing reliance on centralized storage of personal video data and improving generalization across diverse environments. Grad-CAM++ is incorporated to provide clear and intuitive visual explanations of the model's decisions, supporting transparency, trustworthiness, and responsible AI principles. Fairness evaluation is also integrated to ensure that the system performs consistently across different user groups and fall types. The overall system workflow proceeds as follows:

1. Video streams are captured from surveillance cameras.
2. MobileNetV2 extracts spatial features from individual frames.
3. The GRU processes the sequence of extracted features to learn temporal motion patterns.
4. During training, federated learning aggregates locally trained models from multiple clients while keeping the raw video data on the local devices.
5. Fairness and explainability components assess the model's reliability and responsible behavior.
6. During testing, a sliding-window mechanism classifies each video segment as fall or non-fall, and a fall alert is triggered when two consecutive windows are detected as falls.

The following subsections describe each module in detail, along with the federated training process and the proposed inference strategy.

3.1. Spatial feature extraction with MobileNet

MobileNetV2 serves as the spatial feature extraction backbone in the proposed fall detection framework, providing a lightweight and computationally efficient architecture for extracting spatial features from video frames. Its core building blocks are inverted residual blocks with linear bottlenecks, where each block first expands the input channels using a 1×1 convolution, applies a 3×3 depthwise convolution to capture spatial patterns separately on each channel, and then projects the features back to a compact representation using another 1×1 convolution (pointwise) (Sandler et al., 2018). The use of depthwise separable convolutions—splitting the spatial filtering (depthwise) and channel combination (pointwise)—dramatically reduces the number of parameters and floating-point operations compared to standard convolutions, enabling faster inference with minimal loss of accuracy. MobileNetV2 contains 17 inverted residual blocks, plus initial and final convolutions, totaling 53 layers, including all expansion, depthwise, and projection layers. Each block preserves essential feature information while keeping computations low, generating compact and informative spatial feature maps that encode body posture and scene context. These features are then passed to the temporal GRU module, which models motion dynamics across consecutive frames to accurately distinguish falls from normal daily activities.

3.2. Temporal modeling with GRU

Since, MobileNet only processes each frame independently and does not connect information across frames, it can miss important temporal patterns in videos. Actions like falling happen over several frames, and analyzing a single frame alone is often insufficient to accurately detect the action. To address this limitation, the proposed technique adds a GRU after MobileNet to capture the temporal dependencies between consecutive frames.

In this approach, each video frame I_t , where t denotes the current frame index in the video sequence, is first passed through MobileNet to extract a feature vector $x_t = f_{\text{MobileNet}}(I_t)$, which encodes spatial information such as body posture and orientation. The sequence of features $x_1, x_2, \dots, x_T$, where T represents the total number of frames in the video, are then fed into a GRU, which is a type of RNN designed to capture long-term dependencies efficiently. The GRU architecture consists of a hidden state h_t and two gating mechanisms: the update gate z_t and the reset gate r_t . These gates control how information is passed along the sequence, enabling the GRU to remember relevant temporal information while ignoring irrelevant details. At each time step, the GRU updates its hidden state using the following equations (Dey & Salemt, 2017):

$$\begin{aligned} z_t &= \sigma(W_z x_t + U_z h_{t-1}) \text{ (update gate)} \\ r_t &= \sigma(W_r x_t + U_r h_{t-1}) \text{ (reset gate)} \\ \tilde{h}_t &= \tanh(W_h x_t + U_h (r_t \odot h_{t-1})) \text{ (candidate hidden state)} \\ h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \text{ (final hidden state at time step } t) \end{aligned}$$

Where: σ = sigmoid function, $\odot$ = element-wise multiplication and W and U = weight matrices

During training, the GRU updates its parameters through Back-propagation Through Time (BPTT), using the loss between predicted and true labels to learn temporal patterns that distinguish abrupt fall movements from gradual or routine daily activities.

To support efficient inference, the model operates on the video stream using a sequential, non-overlapping sliding-window approach. Each window consists of 20 consecutive frames ($F_1, F_2, \dots, F_{20}$), from which five frames are uniformly sampled (e.g., $F_1, F_5, F_{10}, F_{15}, F_{20}$). These sampled frames provide a compact temporal summary while reducing computational overhead and redundancy. During inference, each sampled sequence is passed through the GRU, where the hidden state h_t evolves based on learned motion patterns. Sudden posture change, such as rapid descent or collapse, lead to substantial updates in h_t , while smoother motions such as bending or walking produce gradual state transitions. The reset gate r_t helps discard stale information when motion abruptly changes, while the update gate z_t determines how much new information should override existing memory, enabling the network to track the motion trajectory effectively.

After processing each window, the final hidden state is passed to the classifier to produce a prediction $P_w \in \{\text{fall}, \text{non-fall}\}$. To minimize false alarms caused by short transitional movements, a decision rule is applied whereby a fall is confirmed only if two consecutive windows are both predicted as fall:

$$\text{Fall} = \begin{cases} 1, & \text{if } P_t = \text{fall and } P_{t-1} = \text{fall} \\ 0, & \text{otherwise} \end{cases}$$

3.3. Federated learning for decentralized training

In video-based elderly monitoring applications, multiple cameras may be deployed to enable continuous observation and timely fall detection. Traditionally, video streams are transmitted to a centralized server for data processing and model training. However, transferring and storing continuous high-resolution video from multiple devices can create significant challenges related to data management, bandwidth consumption, and storage requirements. Furthermore, in healthcare and assisted-living facilities where numerous individuals are monitored simultaneously, these challenges become increasingly pronounced. Centralized processing also requires large volumes of video data to be collected and maintained at a single location, increasing reliance on centralized storage and potentially limiting scalability across diverse

deployment environments. In addition, handling large video datasets may increase network traffic, storage costs, and compliance requirements associated with managing personal healthcare data. These challenges motivate the development of decentralized learning frameworks that support collaborative model training without requiring centralized collection of raw video recordings.

To address these limitations, the proposed system adopts a simulated federated learning framework (Mammen, 2021), enabling collaborative training of fall detection models across multiple video sources while reducing reliance on centralized storage of sensitive video data. The architecture comprises one central server and five client nodes, each representing a distinct camera view or dataset partition, as shown in Fig. 1. Each client trains a local model on its data and transmits only updated parameters to the central server for aggregation. This decentralized training strategy eliminates the need to transfer raw video data to a central repository and enables learning from diverse distributed data sources. The federated learning workflow consists of the following main steps:

Step 1: Global Model Initialization:

The central server initializes a global fall detection model and shares the initial parameters with all client nodes.

Step 2: Model Distribution:

The initialized model parameters are distributed to five clients — each representing a unique data source.

- Four clients correspond to distinct camera views from the Le2i dataset.
- One client represents the CAUCA dataset, treated as a single entity due to its consistent environment and camera configuration.

Step 3: Local Training:

Each client trains the model locally using its own dataset segment. Raw video data remain on the client device, reducing reliance on centralized storage and lowering data transmission requirements.

Step 4: Model Updates:

Clients send only the updated model parameters (weights) to the

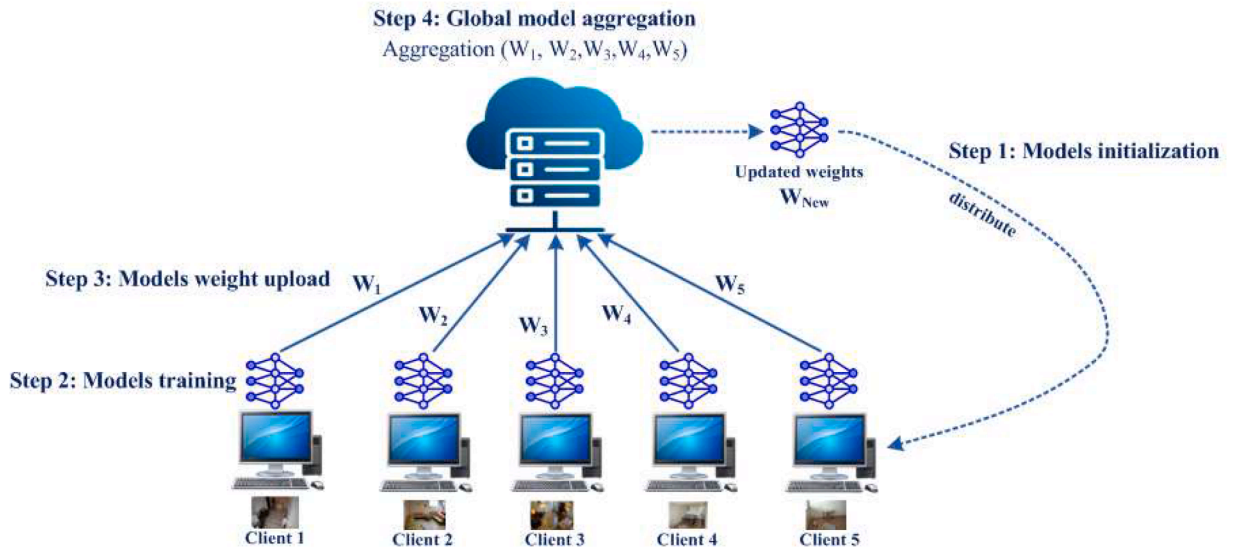


Fig. 1. The proposed federated learning workflow.

central server after local training.

Step 5: Aggregation:

The central server aggregates the received model updates using a weighted averaging method (e.g., FedAvg) to update the global model.

Step 6: Iteration:

Steps 2–5 are repeated until convergence is achieved, improving model performance across diverse environments.

It should be noted that the proposed federated learning framework does not incorporate differential privacy mechanisms and therefore does not provide formal privacy guarantees. While raw video data remain on local devices and are not transmitted to the central server, potential privacy threats such as gradient inversion, membership inference, and model reconstruction attacks may still exist. Consequently, the federated learning component should be viewed as a decentralized training strategy rather than a formally privacy-preserving mechanism.

3.4. Explainability

Misclassifications in fall detection systems for elderly monitoring can lead to serious real-world consequences. False positives, when a system incorrectly identifies a fall, may cause unnecessary alarms, caregiver fatigue, and inefficient use of emergency resources, whereas false negatives, when a true fall is missed, can delay medical response and increase the risk of injury or fatal outcomes. These limitations arise because deep learning models often behave as black boxes, providing accurate outputs without revealing the reasoning behind their predictions.

To overcome this challenge, the proposed research integrates XAI techniques within the proposed fall detection framework based on MobileNetV2 and a GRU network. Although this hybrid architecture

achieves strong detection accuracy, its underlying reasoning remains opaque. Therefore, Grad-CAM++ is applied to enhance interpretability by visually highlighting the image regions that most influence the model's fall or non-fall predictions.

In the proposed model, Grad-CAM++ generates Class Activation Maps (CAMs) that assign importance scores to different spatial regions in the final convolutional layers of MobileNetV2. Grad-CAM++ uses the last convolutional layer because it captures high-level semantic features while preserving spatial information, unlike earlier layers that focus only on low-level details (Chattopadhyay et al., 2018).

The generated heatmaps are then overlaid on the original video frames to indicate which body parts or movements contributed most to the classification decision. This visualization provides valuable transparency, allowing developers and caregivers to verify that the system bases its predictions on relevant visual cues, thereby improving trust, accountability, and safety in elderly fall detection applications. The Grad-CAM++ process within the proposed MobileNetV2–GRU framework, as shown in Fig. 2, proceeds as follows:

1. Forward propagation: Each video frame I is passed through the MobileNetV2 backbone to extract feature maps from the last convolutional layer. These feature maps are denoted as A^k , where k refers to one specific feature map inside a convolutional layer. Also, the pre-softmax score (logit) for the target class c (y^c) is computed.
2. Gradient computation: For each feature-map channel k and spatial location (i, j) , the partial derivative of y^c with respect to the activation $A_{i,j}^k$ is computed.

$$\frac{\partial y^c}{\partial A_{i,j}^k}$$

Then, compute higher-order derivatives

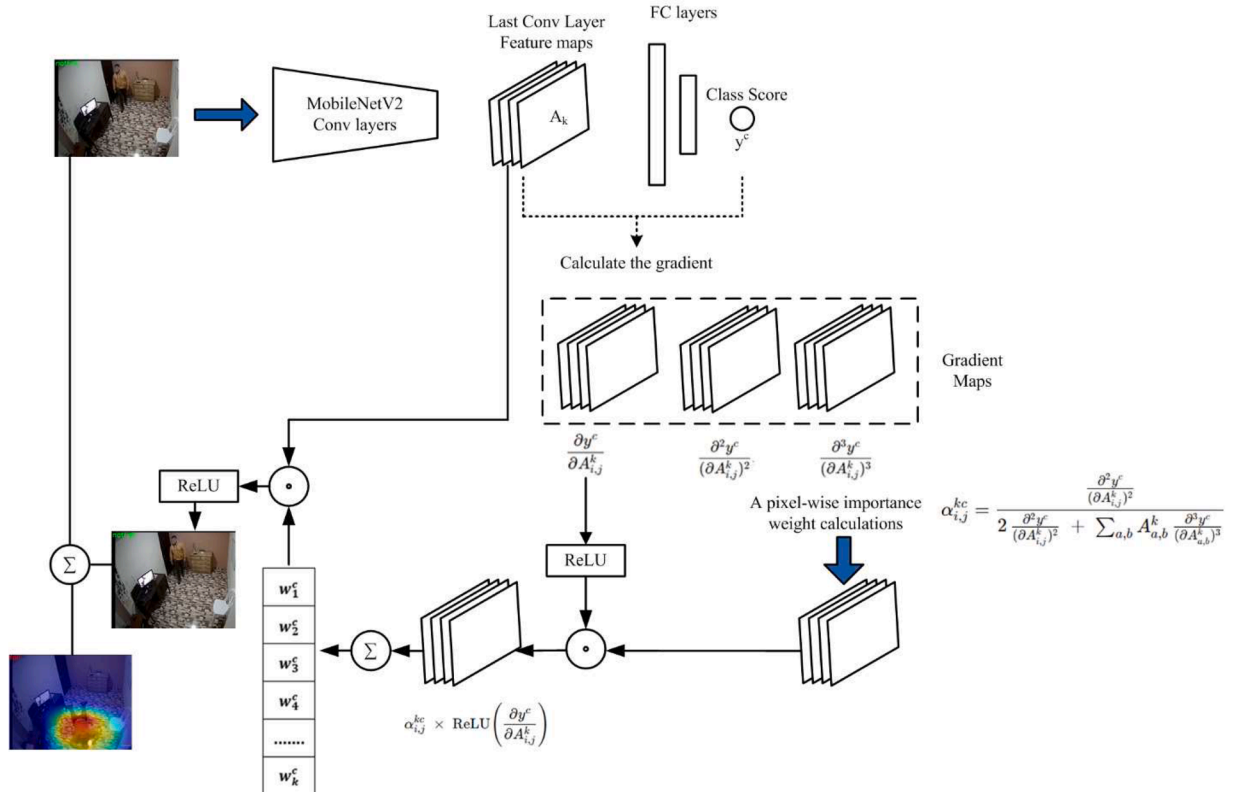


Fig. 2. Grad-CAM++ based visual explanation of the proposed fall detection model decisions.

$$\frac{\partial^2 y^c}{(\partial A_{ij}^k)^2}, \frac{\partial^3 y^c}{(\partial A_{ij}^k)^3}$$

3. Weight calculation: Grad-CAM++ computes pixel-wise importance weights using the positive partial derivatives of the gradients, allowing multiple relevant regions to contribute to the explanation. These pixel-wise weights for each spatial location are calculated as:

$$\alpha_{ij}^{kc} = \frac{\frac{\partial^2 y^c}{(\partial A_{ij}^k)^2}}{2 \frac{\partial^2 y^c}{(\partial A_{ij}^k)^2} + \sum_{a,b} A_{a,b}^k \frac{\partial^3 y^c}{(\partial A_{a,b}^k)^3}}$$

These α coefficients capture location-specific importance, allowing the method to highlight multiple spatial regions that contribute to the class.

4. Channel weights (w_k^c): Aggregate the pixel-wise contributions for each channel and this retains only the positive contributions (via ReLU), aligning the visualization with features that support the predicted class.

$$w_k^c = \sum_i \sum_j \alpha_{ij}^{kc} \times \text{ReLU}\left(\frac{\partial y^c}{\partial A_{ij}^k}\right)$$

5. Class activation map generation: A weighted combination of the feature maps produces the heatmap highlighting the most influential regions for the predicted class. This produces a coarse heatmap at the feature-map resolution, highlighting the spatial areas most relevant to class c .

$$L_{\text{Grad-CAM++}}^c = \text{ReLU}\left(\sum_k w_k^c A^k\right)$$

6. Visualization and interpretation: Map the heatmap $L_{\text{Grad-CAM++}}^c$ to the same dimensions as the input frame I and then overlay it to visually indicate the important regions.

4. Experiments and results

This section details the experimental setup and the datasets employed in the study, followed by an evaluation of the proposed model's performance across its components. The findings are further compared with state-of-the-art models to highlight the strengths of our approach.

4.1. Datasets

4.1.1. CAUCAFall

The CAUCAFall dataset, developed by Guerrero et al. (2022), provides a diverse and realistic collection of fall and non-fall activities recorded in an uncontrolled home environment. It features ten participants, five men and five women, who vary in age, height, weight, and clothing style. The recordings were captured using a standard RGB camera at 23 frames per second and a resolution of 1080×960 pixels, resulting in approximately 8 GB of video data. Each subject performed five types of falls—forward, backward, left-side, right-side, and from a sitting position—and five daily living activities, including walking, hopping, object retrieval, sitting, and kneeling. The videos reflect natural variability in lighting (daylight, artificial, and night conditions),

occlusions, background motion, and differences in camera angles, distances, and room textures, making the dataset highly representative of real-world home scenarios. Fig. 3(a) shows sample frames from the CAUCAFall dataset, illustrating different body postures and fall stages captured in a realistic home environment.

4.1.2. Le2i

The Le2i fall detection dataset (L. D. V. Tuyen, 2022) provides 221 video sequences, containing 131 simulated falls and 90 Activities of Daily Living (ADL). Each sequence is recorded with a fixed camera at a frame rate of 25 fps and a resolution of 320×240 pixels. Different actors perform the activities in four environments: home, office, coffee room, and lecture room. Out of the total videos, 191 are accompanied by manual annotations that supply ground-truth details on fall locations and body positions across the image frames. Fig. 3(b) presents sample frames from the Le2i dataset, recorded in controlled indoor settings across different rooms.

4.2. Experimental setup

The proposed model is trained and evaluated using videos from two benchmark fall detection datasets, CAUCAFall and Le2i, with an 80:20 train-test split. Rather than using each dataset independently, both datasets are integrated into a simulated federated learning framework to create a heterogeneous client distribution. The Le2i dataset, which includes recordings from four different indoor rooms, is assigned to four clients, while one client represents the CAUCAFall dataset, which contains videos captured under different recording conditions. This client partition introduces diversity in scene characteristics, including background, lighting, camera viewpoints, and recording environments, allowing the model to learn from heterogeneous data sources rather than from a single dataset. It should be noted that this client configuration represents a proof-of-concept simulation and does not reflect a real-world federated deployment with independently collected non-IID data from geographically distributed institutions or homes.

Within this federated setup, the proposed architecture combines MobileNetV2 as a lightweight feature extraction backbone with a GRU layer to capture temporal dependencies between frames. The MobileNetV2 component is initialized with pre-trained weights to improve convergence and feature quality, while the GRU layer consists of one layer with a hidden size of 256. Model training is optimized using the Adam optimizer with a learning rate of 1×10^{-4} and Cross-Entropy Loss as the objective function. During federated training, each client conducts one local epoch in every communication round. The global model is then updated through 20 rounds of parameter aggregation, with a batch size of 4 employed for local training. All experiments were conducted on a laptop equipped with an 11th Generation Intel Core i7 processor running at 2.3 GHz, 32 GB of RAM, and an NVIDIA GeForce RTX 3060 Laptop GPU with 6 GB of dedicated VRAM. The computational performance and inference latency reported in this study are therefore specific to this hardware configuration. Performance on resource-constrained edge devices, such as Raspberry Pi systems, embedded processors, and mobile AI accelerators, was not evaluated in the current study.

4.3. Simulation results

4.3.1. Federated learning convergence analysis

Fig. 4 present the test set accuracy and loss of the proposed model after each federated training round. Training is performed on long video sequences (approximately 200 frames each), from which 20 frames are sampled for participation in the training phase. The federated learning process is conducted over 20 communication rounds, with each client performing one local training epoch per round using a batch size of 4 and a learning rate of 1×10^{-4} . After each round, the locally updated model weights are sent to the central server, where they are aggregated and averaged to form the updated global model. The global model is

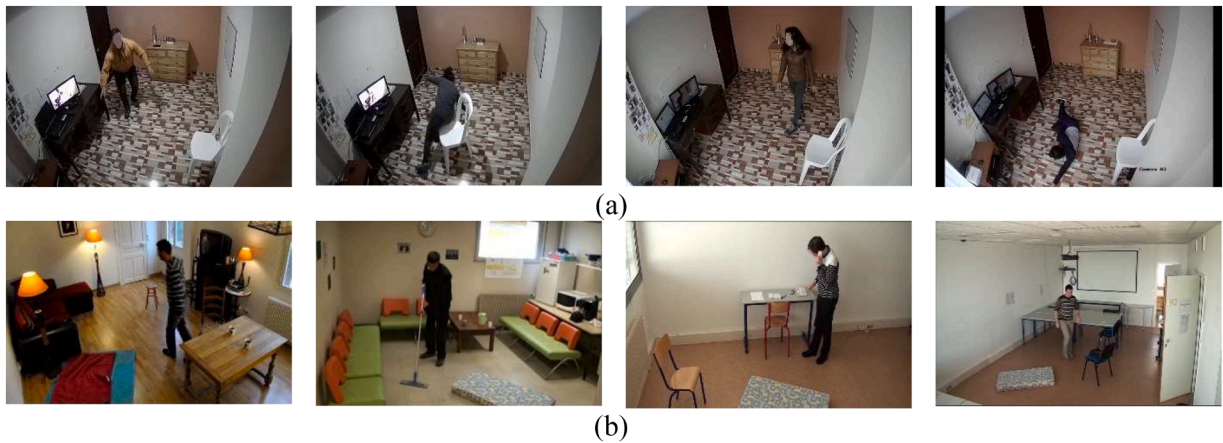


Fig. 3. Sample frames from the two datasets used in this study: (a) CAUCAFall and (b) Le2i.

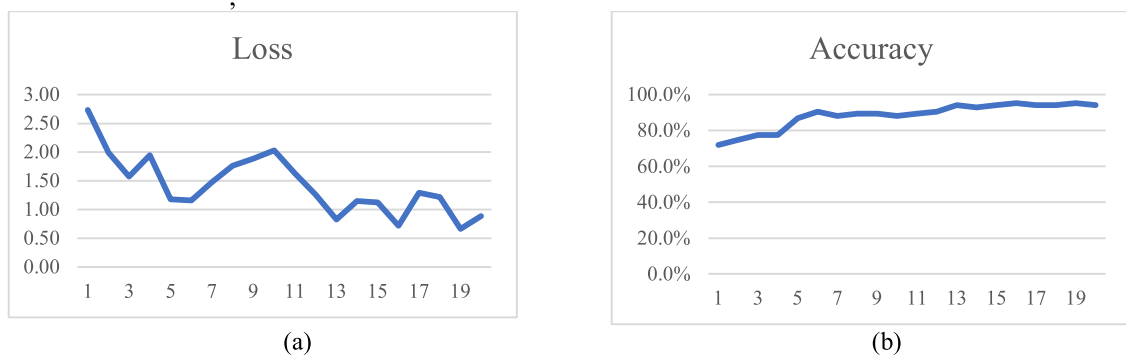


Fig. 4. Training accuracy and loss of the proposed model across 20 federated training rounds.

then evaluated on the test set at the server side to compute the corresponding accuracy and loss values for that round. As shown in Fig. 4, the accuracy curve displays a steady upward trend, indicating progressive improvement of the global model as knowledge from distributed clients is integrated. The accuracy stabilizes after approximately the 15th round, suggesting convergence, while the loss curve continues to decrease smoothly, confirming effective optimization. Among all rounds, the model obtained at round 19 achieves the highest accuracy and lowest loss, and is therefore selected as the best-performing global model. The relatively small number of communication rounds required for convergence demonstrates the efficiency of the proposed federated

learning framework and helps reduce communication overhead during training. Moreover, the smooth behavior of both curves reflects the robustness of the proposed MobileNetV2-GRU architecture and the stability of the chosen hyperparameters, even under non-identically distributed client data.

To further assess the effectiveness of the proposed federated learning strategy, an additional experiment is conducted using a centralized training configuration in which all samples from the CAUCA Fall and Le2i datasets are pooled and used to train the same MobileNetV2-GRU architecture. Fig. 5 compares the accuracy and loss obtained under centralized and federated learning settings. As expected, centralized

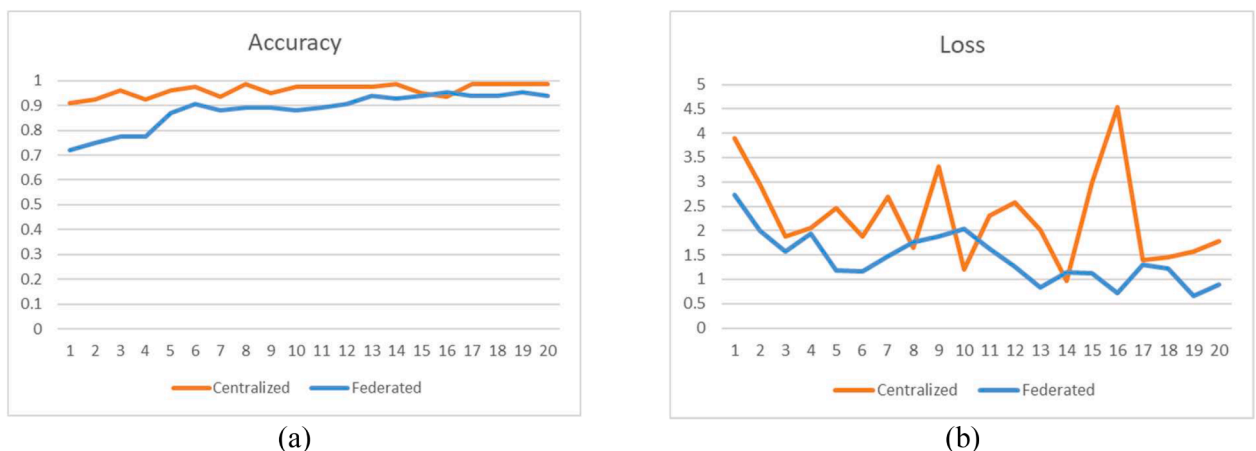


Fig. 5. Performance comparison between centralized and federated learning using the same MobileNetV2-GRU architecture and training data.

training achieves slightly higher performance, outperforming the federated model by 3.73 percentage points in terms of final accuracy. This difference can be attributed to the availability of all training samples at a single location, allowing the model to directly optimize over the complete data distribution. Nevertheless, the federated model maintains competitive performance while enabling decentralized training across multiple clients. Although federated learning introduces a modest reduction in accuracy, it eliminates the need to continuously transfer and centrally store large volumes of raw video recordings. Instead, only model parameters are exchanged between clients and the server, substantially reducing centralized storage requirements and data transmission overhead. Furthermore, keeping video data at the source supports decentralized data management, minimizes exposure of sensitive personal recordings to centralized repositories, and reduces the risk of unauthorized access, disclosure, or misuse of video data. Therefore, the observed performance gap represents a reasonable trade-off between predictive performance and the practical advantages of decentralized model training in healthcare and assisted-living environments.

While the convergence results demonstrate the effectiveness of the proposed federated learning strategy under the considered experimental conditions, it should be noted that the federated learning configuration employed in this study represents a proof-of-concept simulation rather than a fully deployed real-world federated environment. The five clients are created by partitioning the publicly available CAUCA Fall and Le2i datasets into separate local datasets. While this setup demonstrates the feasibility of decentralized training and model aggregation, it does not fully capture practical deployment challenges such as geographically distributed homes, heterogeneous camera hardware, varying network conditions, client dropout, and naturally occurring non-identically distributed (non-IID) data. Therefore, the reported results should be interpreted as an initial validation of the proposed federated learning framework. Future work will investigate deployment across distributed healthcare environments to further assess scalability, communication efficiency, and robustness.

4.3.2. Computational and deployment analysis

To assess the inference efficiency of the proposed framework, a detailed time-cost analysis is conducted on a representative test video containing 4204 frames with a duration of 140.13 s. Table 1 presents the latency breakdown of the main computational components. MobileNetV2 feature extraction requires 36.22 ms and accounts for 96.8% of the total inference time, representing the primary computational cost of the framework. In contrast, the GRU temporal modeling module requires only 0.93 ms (2.5% of total inference time), while the fully connected classification layer contributes 0.20 ms (0.5%). It can be noticed that the GRU introduces negligible computational overhead because it operates on compact feature representations generated by MobileNetV2 rather than directly on raw image frames. Furthermore, the relatively short temporal sequence used in this work substantially reduces the computational burden of temporal modeling.

Beyond inference efficiency, the communication overhead associated with federated training is also an important deployment consideration. The proposed federated learning framework requires the exchange of model parameters rather than raw video data. The final MobileNetV2-GRU model has a disk footprint of approximately 13.5 MB. During each communication round, every client downloads the

global model and uploads the locally updated model, resulting in approximately 27 MB of communication per client per round. Across the 20 federated training rounds used in this study, the total communication cost is approximately 540 MB per client. Although federated learning introduces communication overhead through model exchange, this cost is incurred only during training and remains substantially lower than continuously transmitting and centrally storing raw video recordings. It should be noted that the current study assumes full client participation throughout all federated training rounds. Consequently, the impact of client dropout and communication failures on model convergence and performance is beyond the scope of the present work and was not evaluated.

4.3.3. False alarm analysis

False fall alarms may occur in activities that visually resemble falls, such as picking up objects from the floor or kneeling as illustrated in Fig. 6. In these cases, the system may temporarily classify the movement as a fall due to the rapid downward motion of the upper body or a sudden change in posture. However, when the individual returns to an upright position, the prediction stabilizes, and the false alarm ceases. To further minimize these errors, the system confirms a fall event only when two consecutive windows are both classified as falls. This rule significantly enhances detection reliability by filtering out short or ambiguous actions that do not correspond to real fall events. Nevertheless, some daily activities, such as kneeling or bending for a prolonged period, may extend across more than two windows, occasionally leading to false alarms, as their motion patterns closely resemble genuine falls. These cases highlight the difficulty of differentiating between intentional and accidental downward movements in video-based systems. Despite these limitations, the proposed temporal decision strategy effectively reduces the frequency and persistence of false alarms, maintaining a strong balance between sensitivity to actual falls and robustness against normal daily activities.

To quantitatively assess the impact of false detections during continuous monitoring, an additional false alarm analysis is conducted using long-duration video sequences from the realistic fall dataset of Baldewijns et al. (2016). During the 7 h and 34 min of continuous monitoring, the proposed framework generates three false alarms, corresponding to approximately 0.40 false alarms per hour. Most false detections occur in challenging scenarios, including kneeling, bending, and walking activities that occasionally resemble fall-like body postures or motion patterns as shown in Fig. 7. Although false alarms may occasionally require additional verification by caregivers, they are generally preferable to missed fall events, which may leave an injured individual unattended for an extended period. These results provide an initial assessment of the framework's behavior under continuous monitoring conditions and demonstrate its ability to maintain a relatively low false-alarm rate while preserving high sensitivity to fall events. However, the current evaluation is conducted on benchmark data rather than real-world residential deployments. Therefore, further validation through long-term monitoring in healthcare and assisted-living environments remains necessary to fully assess false-alarm rates and practical deployment performance.

4.3.4. Explainability analysis

To enhance the interpretability of the proposed model, several XAI methods are examined, including GradCAM (Aminudin et al., 2025), GradCAM++ (Chattopadhyay et al., 2018), EigenCAM (Rahman et al., 2022), EigenGradCAM (Tang et al., 2024), Feature Explanation Map (FEM) (Pan et al., 2020), FinerCAM (Zhang et al., 2025), HiResCAM (Draeos & Carin, 2020), and XGradCAM (Fu et al., 2020). As presented in Fig. 8, GradCAM++ produces the most accurate and meaningful visualization results. Compared with the original GradCAM, which generates broader and less focused activation maps, GradCAM++ provides sharper localization and clearer attention on the regions associated with the falling action. This improvement results from its weighted

Table 1

Inference latency breakdown of the proposed MobileNetV2-GRU framework.

Component	Latency (ms)	% of Inference
MobileNetV2	36.22	96.8
GRU	0.93	2.5
FC Classifier	0.20	0.5
Total Inference	37.43	100

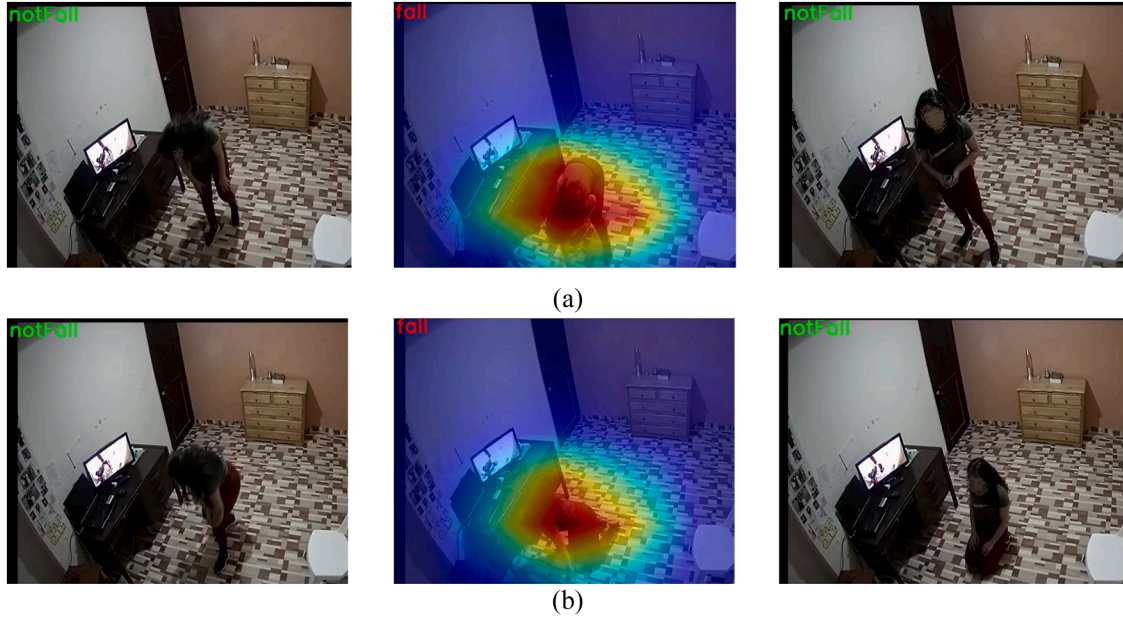


Fig. 6. Example of false fall alarms generated during two daily activities: (a) picking up an object and (b) kneeling.



Fig. 7. Representative false alarm cases during continuous monitoring, including kneeling, bending, and walking activities. The lower row shows the corresponding Grad-CAM++ visualizations.

gradient computation, which enables the identification of multiple discriminative areas simultaneously. Such behavior is particularly beneficial for video-based fall detection, where movements involve several body parts. GradCAM++ also demonstrates stable and consistent explanations across consecutive frames, maintaining focus on the human motion rather than the background. In contrast, EigenCAM and HiResCAM often highlight irrelevant areas, while FEM produces scattered activations. Based on these observations, GradCAM++ is selected as the most suitable XAI technique, as it delivers clear, consistent, and contextually relevant visual interpretations of the model's decisions.

4.3.5. Fairness evaluation

When a model is unfair or biased, it may misclassify certain groups more frequently, leading to unequal safety outcomes, such as delayed detection for one gender or fall type. So, this raises the importance of responsible AI when designing AI applications to guarantee the model output is explainable and fair. So, in this paper, the model performance predictions are evaluated to ensure that the system performs equitably across all users, regardless of factors such as gender, age, or fall direction. Therefore, by assessing fairness alongside accuracy, developers can

identify hidden biases, improve model reliability, and promote trust and accountability in the proposed model. To support this objective, Fairlearn (an open source library) (Fairlearn) is used in the proposed system to assess the fairness of the model. The fairness is evaluated through two aspects: gender (male and female) and fall direction (back, front, side, and from a chair). Fig. 9 presents the Fairlearn evaluation of model performance across the fall directions. It can be noticed that Fairlearn results show consistent F1, recall, and precision across all fall directions, indicating that the model performs fairly and accurately without bias toward any specific fall type (Beya et al., 2015).

Moreover, the model predictions are evaluated across the gender of the fallen persons, and the results are shown in Fig. 10. The results show nearly identical F1, recall, and precision scores for male and female subjects, indicating that the fall detection model performs equitably across genders without observable bias.

Furthermore, Fig. 11 presents the combined fairness evaluation of the proposed fall detection model across both gender and fall types. This figure examines the interaction between the two factors to determine whether performance disparities appear in specific combinations, such as female-side or male-chair falls. The results show that the model

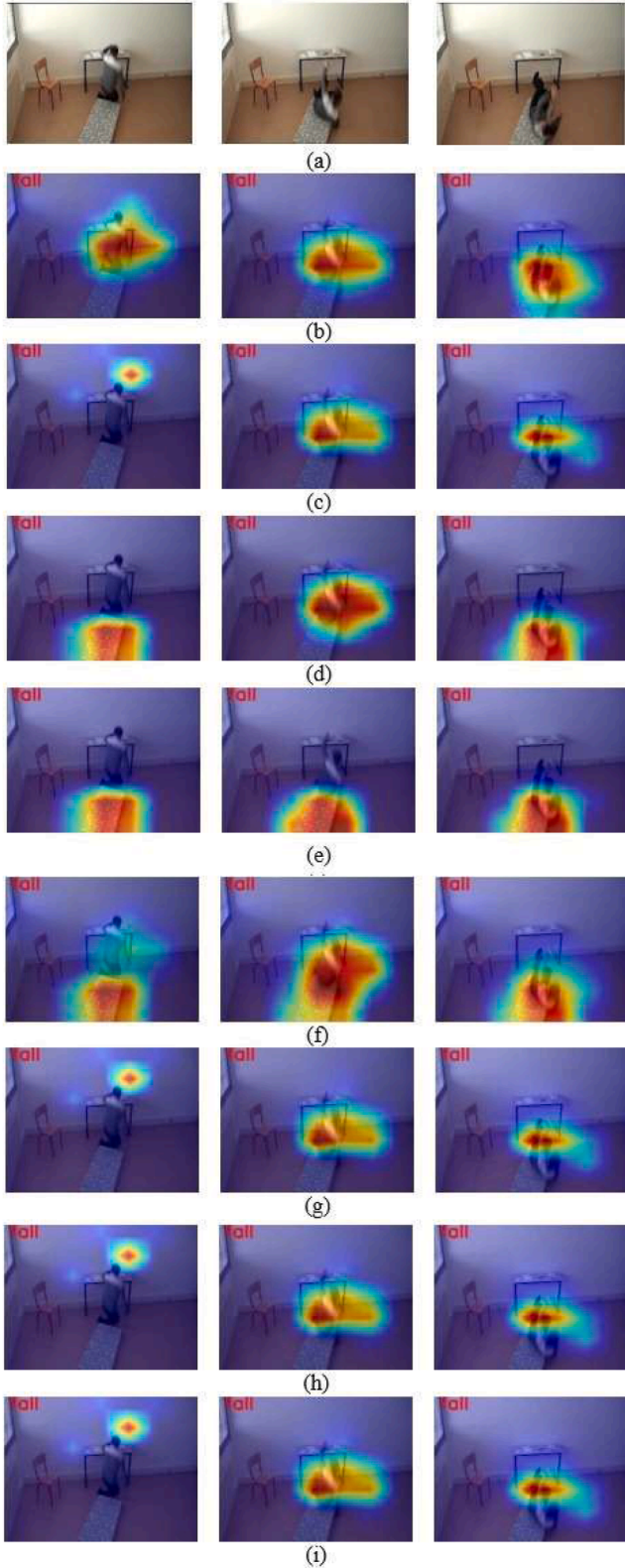


Fig. 8. Visualization results of different XAI methods used to interpret the proposed fall detection model. (a) original frames (b) GradCAM++, (c) Grad-CAM, (d) EigenGradCAM, (e) EigenCAM, (f) FEM, (g) FinerCAM, (h) HiResCAM, (i) XGradCAM.

maintains consistently high F1, recall, and precision values across all gender-fall type pairs, confirming that it performs uniformly well regardless of the subject's gender or the nature of the fall. This combined analysis provides stronger evidence of the model's fairness and robustness, demonstrating that no intersectional bias arises when both demographic and behavioral variables are considered simultaneously. Such findings further reinforce the reliability of the proposed system for real-world use, where both gender diversity and variability in fall dynamics are expected.

To further evaluate the impact of fairness-aware partitioning, the federated model was tested using random and fairness-aware data splits. Random splits showed considerable performance variation, with accuracies ranging from 64.87% to 97.37% (mean: $85.19\% \pm 12.9\%$). Fairness-aware splits, evaluated across two runs, achieved 95.00% and 95.20%, both within the upper range of the random split results and without comparable performance degradation. The 95.20% fairness-aware model is selected as the final model and integrated with the window-based fall detection mechanism, achieving an overall accuracy of 97.14%. Although evaluated on a limited number of fairness-aware runs, these results suggest that fairness-aware partitioning can reduce the risk of performance drops caused by random splitting while maintaining detection accuracy.

4.4. Comparison with the conventional methods

Table 2 compares the proposed method with several existing fall-detection approaches in terms of accuracy, precision, recall, and F1-score. Overall, the proposed model demonstrates competitive performance compared with state-of-the-art methods while achieving a favorable balance between accuracy, precision, recall, and F1-score. For example, compared to [Romaissa et al. \(2020\)](#), it achieves notable gains of +12.54% in accuracy, +9.26% in precision, +6.75% in recall, and +7.99% in F1-score, reflecting the strength of modeling full spatio-temporal motion patterns rather than depending on handcrafted posture features. Similarly, when compared with [Tran et al. \(2025\)](#), the proposed method improves by +4.95% in accuracy, +17.26% in precision, +12.75% in recall, and +14.99% in F1-score, demonstrating robustness against key-point detection errors and occlusions.

Furthermore, the proposed method shows clear advantages over deep learning approaches trained on individual datasets, highlighting the benefits of federated training across heterogeneous datasets. For instance, compared to [Alam et al. \(2025\)](#), it improves accuracy by 3.47%, precision by 5.84%, and F1-score by 4.04%. In contrast, compared to [Oliu \(2025\)](#), it shows a slight decrease of 1.36% in accuracy; however, Marc Oliu's higher accuracy of 98.50% comes at the expense of lower precision, as the proposed method attains 99.26%, minimizing false positives, a key factor for reducing alarm fatigue and maintaining user trust. Additionally, training on the combined CAUCA and Le2i datasets allows the model to learn generalizable representations that perform consistently across diverse real-world scenarios, rather than overfitting to a single dataset.

When evaluated against [Gao et al. \(2023\)](#), which combines OpenPose skeletal keypoint extraction with a lightweight MobileNetV2 architecture and achieves 98.6% accuracy on Le2i, the proposed method demonstrates comparable accuracy of 97.14%, with slightly higher precision (99.26% vs. 98.65%) and slightly lower recall (96.75% vs. 97.7%). While Mengqi Gao et al.'s two-stage approach performs well, it faces limitations such as higher latency from sequential OpenPose extraction and MobileNetV2 classification, strong dependence on OpenPose accuracy, and reduced robustness in cluttered environments, which can propagate errors.

Similarly, compared to [Zou et al. \(2021\)](#), which uses movement tube detection with 3D CNNs and achieves 97.23% accuracy on Le2i, the proposed method shows a minimal decrease in accuracy (0.09 percentage points). However, Song Zou's approach suffers from high computational cost, reliance on accurate frame-by-frame person

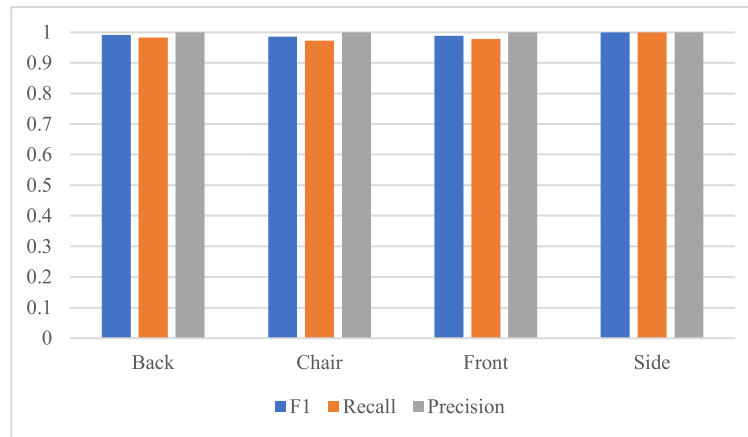


Fig. 9. Fairlearn evaluation of model performance across fall directions.

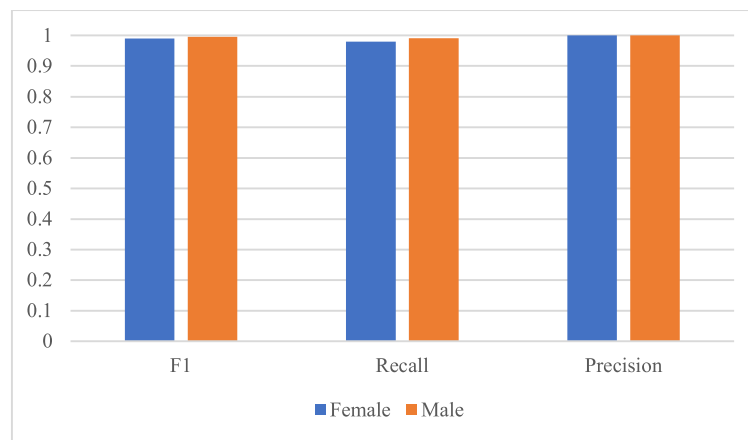


Fig. 10. Model fairness evaluation across gender groups using Fairlearn.

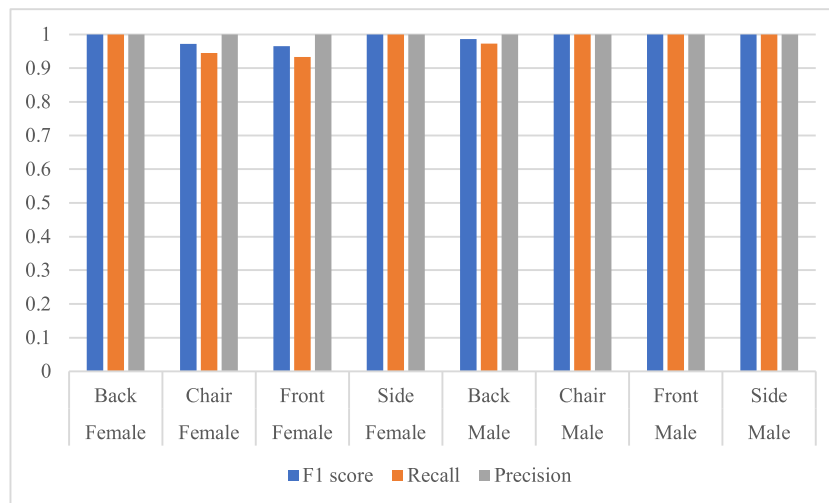


Fig. 11. Combined evaluation of fairness across gender and fall type using F1 score, Recall, and Precision.

detection, and sensitivity to partial occlusions. Against [McCall et al. \(2024\)](#), evaluated on CAUCA Fall data with 97.95% accuracy, the proposed method reaches 97.14% accuracy; however, the lack of precision, recall, and F1-score metrics in their report prevents a full multi-metric comparison.

In comparison with Wang et al. ([Wang & Lin, 2024](#)), which achieves

98.86% accuracy, 90.02% precision, and 90.50% F1-score on Le2i, the proposed method shows slightly lower accuracy (97.14%) but substantially higher precision (99.26%) and F1-score (97.99%). This indicates that, although Chang-Yueh Wang et al.'s method achieves higher overall accuracy, the proposed method is more effective at minimizing false positives and maintaining a better balance between precision and recall.

Table 2

Quantitative performance comparison between the proposed model and state-of-the-art methods.

Authors	Dataset	Accuracy	Precision	Recall	F1-Score
Wang et al. (Wang & Lin, 2024)	Le2i	98.86	90.02	-	90.50%
Romaissa et al. (2020)	Le2i	84.60%	90.00%	90.00%	90.00%
Oliu (Oliu, 2025)	CAUCA	98.50%	97.55%	99.08 %	-
Alam (Alam et al., 2025)	CAUCA	93.67%	93.42%	-	93.95%
Gao et al. (2023)	Le2i	98.6%	98.65%	97.7%	-
McCall et al. (2024)	CAUCA	97.95	-	-	-
Zou et al. (2021)	Le2i	97.23%	-	-	-
Zaghden et al. (2025)	CAUCA	99.6%	99.72%	99.93%	99.82%
Tran et al. (2025)	CAUCA	92.19%	82.00%	84.00%	83.00%
Proposed method	CAUCA & Le2i	97.14%	99.26%	96.75%	97.99%

Finally, Zaghden et al. (2025) achieves the highest reported performance on CAUCA alone, with 99.6% accuracy, 99.72% precision, 99.93% recall, and 99.82% F1-score. The proposed method performs slightly lower, with differences of 2.46, 0.46, 3.18, and 1.83 percentage points, respectively. This performance gap may be attributed to differences in architecture and optimization strategy. Their framework combines Faster R-CNN, OpenPose-based keypoint extraction, GAN-based data augmentation, attention mechanisms, and a modified YOLOv8 detector, providing detailed posture and localization information that may improve discrimination between falls and visually similar activities such as kneeling or bending. However, these improvements are achieved through a substantially more complex pipeline. In contrast, the proposed framework employs a lightweight MobileNetV2–GRU architecture that achieves competitive performance while maintaining high inference efficiency, together with explainability through Grad-CAM++, fairness evaluation using Fairlearn, and decentralized federated learning.

4.5. Ablation study

Ablation experiments are conducted to examine the impact of temporal configuration on both the training and inference stages of the proposed fall detection model. Two complementary studies are performed: (1) an ablation study on training temporal context and (2) an ablation study on window configuration and latency. Together, these experiments aim to identify the temporal setup that best balances accuracy, responsiveness, and robustness.

In the first study, the effect of temporal input length on model performance is evaluated using three configurations: Model 20, Model 40, and the Full Model. Models 20 and 40 are trained using sliding windows of 20 and 40 consecutive frames, respectively, while the Full Model is trained on entire video sequences without segmentation. For fair comparison, all models are tested using 20-frame sliding windows. As shown in Table 3, Model 40 achieves a very high recall for the Fall class (0.99), indicating that nearly all true fall events are detected; however, its

precision drops substantially (0.65), suggesting a high number of false positives. In contrast, Model 20 demonstrates a more balanced performance, with a recall of 0.89 and a precision of 0.72 for the Fall class, capturing shorter fall events effectively but missing longer temporal patterns. The Full Model outperforms both window-based models across all metrics, maintaining high precision and recall for both Fall and Not Fall classes (precision: 0.92–0.99, recall: 0.97–0.98), even when evaluated on 20-frame windows. This indicates that exposure to complete motion sequences during training enables the model to learn richer temporal patterns and contextual cues, allowing it to accurately recognize partial motion sequences at inference.

These results in Table 3 highlight the importance of incorporating global temporal information during training. While longer sliding windows improve sensitivity, training on full video sequences provides the best trade-off between precision and recall, yielding the most robust and generalizable configuration across the evaluated benchmark datasets.

To optimize both accuracy and computational efficiency, additional experiments examine the influence of temporal inference settings on the proposed fall detection system. Specifically, these studies explore how window size, step size (number of frames processed per inference), and the number of consecutive windows required to confirm a fall affect performance. These parameters are carefully tuned to balance F1-score with inference latency, defined as the number of frames needed to produce a detection, which is critical for timely intervention. Window size determines the temporal context the model observes, with larger windows capturing longer motion sequences but introducing delays, while step size controls prediction frequency, with smaller steps enhancing responsiveness. Successive window confirmation is employed to filter out transient misclassifications caused by brief non-fall movements, such as bending or sitting.

The results in Table 4 reveal that a 20-frame window, processed 5 frames per step, and double confirmation for fall events achieve the optimal balance between sensitivity and speed. In this configuration, a fall is confirmed only after two consecutive positive windows, whereas non-fall events require a single negative window, allowing the system to respond faster to normal activity while maintaining reliability for fall detection. This setup achieves a high F1-score of 0.98, with inference latencies of 20 frames for non-fall and 40 frames for fall events. Smaller windows reduce latency without sacrificing accuracy, whereas larger windows introduce unnecessary delay, highlighting the importance of carefully balancing temporal context and responsiveness.

Together, these ablation studies underscore the value of global temporal information during training and efficient short-window inference during deployment. By integrating full-sequence training with a 20-frame sliding-window inference strategy and a double-confirmation mechanism, the system achieves a well-balanced synergy of precision, robustness, and rapid decision-making. This configuration enables the model to capture rich temporal patterns during training while maintaining low-latency, high-confidence predictions during deployment. Complementing these design choices, the system achieves an inference throughput of approximately 93 FPS, demonstrating high computational efficiency while maintaining accurate fall detection.

Table 3

Precision and recall of the proposed model for different temporal training configurations.

Model	Class	Precision	Recall
Model 20	Fall	0.72	0.89
	Not Fall	0.95	0.86
Model 40	Fall	0.65	0.99
	Not Fall	0.99	0.79
Model Full	Fall	0.92	0.98
	Not Fall	0.99	0.97

Table 4

Effects of temporal window settings on latency and accuracy.

Window size	Frames per window	Successive windows to decide fall	Latency in a non-fall case	Latency in the fall case	F1 score
40	10	2	40	80	0.978
20	5	2	20	40	0.98
40	10	1	40	40	0.974
20	5	1	20	20	0.975

5. Conclusion

This paper presents a lightweight and explainable fall detection framework that integrates MobileNetV2 for spatial feature extraction, GRU-based temporal modeling, Grad-CAM++ for visual interpretability, and federated learning for decentralized model training. The proposed framework achieves competitive performance on the CAUCA Fall and Le2i benchmark datasets while maintaining high inference efficiency. Experimental results demonstrate the effectiveness of the MobileNetV2-GRU architecture for distinguishing fall events from daily activities, while Grad-CAM++ provides intuitive visual explanations that enhance transparency and trust in the model's predictions. Furthermore, fairness evaluation confirms consistent performance across different fall scenarios and user groups.

The federated learning strategy enables collaborative model training across distributed data sources without requiring centralized collection of raw video recordings. Although centralized training achieves slightly higher accuracy, the federated model maintains competitive performance while reducing reliance on centralized storage and transmission of sensitive personal video data. Computational analysis further demonstrates the efficiency of the proposed framework, and continuous-monitoring experiments provide an initial assessment of false alarms under realistic operating conditions.

Despite these promising results, several limitations remain. The current evaluation relies on publicly available benchmark datasets, while the federated learning framework is assessed through a proof-of-concept simulation. Therefore, additional validation in real-world healthcare and assisted-living environments is necessary to establish the framework's suitability for practical clinical deployment. In addition, false alarms may still occur during challenging daily activities which can exhibit motion patterns similar to falls. Furthermore, while federated learning reduces reliance on centralized storage of raw video data, the proposed framework does not incorporate formal privacy-preserving mechanisms and therefore does not provide formal privacy guarantees. Future work will focus on large-scale validation in real healthcare and assisted-living environments, investigating robustness under heterogeneous deployment conditions, reducing false alarms during daily activities, and integrating advanced privacy-enhancing mechanisms within the federated learning framework. These directions will further strengthen the practical applicability, trustworthiness, and responsible deployment of AI-based fall detection systems.

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Credit author statement

Mona M. Moussa: Conceptualization, Methodology, Software, Investigation, Validation, Writing - review & editing. Rasha Shoitani: Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Writing - original draft, Writing - review & editing. Nahed Tawfik: Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Writing - original draft, Writing - review & editing. All authors contributed to the development of the work, reviewed and approved the final manuscript, and agreed to be accountable for the work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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